



Financial Instrument & Trading

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Commodity Markets:

Physical Goods + Price Risk

Physical market	Financial / derivatives market
Actual crude oil, copper, wheat, gas, coffee, etc.	Contracts whose value depends on commodity prices
Transactions specify quantity, quality, location and delivery date	Futures, options, forwards, swaps
Price is specific to the physical transaction	Usually references a standardized benchmark
Main users: producers, consumers, merchants	Hedgers, traders, funds, market makers

Why does the derivatives market exist?

Existing price risk → **Transfer that risk** → **Someone else takes it**

Copper miner

Owens future copper production

→ Exposed to **falling copper prices**

→ **Sells copper futures**

Cable manufacturer

Needs copper in the future

→ Exposed to **rising copper prices**

→ **Buys copper futures**

Spot



DEFINITION

A **spot transaction** is the purchase or sale of the actual commodity for immediate or near-term delivery under the conventions of that market.

The broader **physical market** also includes transactions for later delivery.

EVERY PHYSICAL TRADE MUST DEFINE

- | | | | |
|----|-----------------|----|---------------|
| 01 | Commodity | 02 | Quality/Grade |
| 03 | Quantity | 04 | Location |
| 05 | Delivery Window | 06 | Price |

$$\text{Physical Price} = \text{Benchmark Price} + \text{Differential}$$

THE DIFFERENTIAL / BASIS REFLECTS

Location & freight

Quality/grade

Storage

Local supply/demand

Delivery timing

Handling/logistics

$$\text{Basis} = \text{Physical Price} - \text{Futures Price}$$

Futures



DEFINITION

A futures contract is a standardized, exchange-traded agreement to buy or sell a specified commodity at a future date

Anatomy of a Futures Contract

Specification	What it tells you
Contract size	Quantity represented by 1 contract
Price quote	Units in which price is displayed
Tick size	Smallest permitted price movement
Contract month	When the contract expires/settles
Settlement	Physical delivery or cash settlement

CONTRACT SNAPSHOT WTI Crude Oil Futures

CONTRACT SIZE **1,000 barrels**

QUOTED IN **\$/barrel**

MINIMUM TICK **\$0.01/barrel = \$10/Contract**

At \$80/bbl **\$80,000 notional**

\$1.00 OIL MOVE
= \$1,000 per contract

How Futures Work in Your Account:

Margin, Leverage & P&L



FUTURES MARGIN ≠ BORROWING

Margin = collateral deposited to support a futures position. It is not a down payment on the commodity, and you are not borrowing the remaining notional value. CME describes futures margin as money that must be deposited and maintained as a performance bond.

MARGIN TERMS

Term	Meaning
Initial margin	Amount required to open the position
Maintenance margin	Minimum account equity required to keep it open
Variation margin	Cash gain/loss generated as the contract is marked to market

EXAMPLE · LONG 1 WTI FUTURE

ASSUME WTI price: \$80/barrel Contract: 1,000 barrels

THEREFORE

\$80 × 1,000 = \$80,000

Notional exposure = \$80,000

Now suppose the required initial margin is \$8,000 purely for illustration.

YOU DEPOSIT

\$8,000

but

YOUR P&L IS CALCULATED ON

1,000 barrels

**This is the source of
futures leverage.**

Options on Futures



OPTIONS ON FUTURES

An option gives the buyer the right, but not the obligation, to take a position in the underlying futures contract at a specified strike price before or at expiration. The seller receives the premium and takes on the corresponding obligation

CALL VS PUT

	Call	Put
Buyer expects	Price ↑	Price ↓
Right to	Buy futures	Sell futures
Maximum buyer loss	Premium	Premium
Used for	Bullish exposure / upside hedge	Bearish exposure / downside hedge

EVERY OPTION HAS 4 CRITICAL INPUTS

01 Underlying future + 02 Strike + 03 Expiration + 04 Premium

Commodity ETFs, ETPs & ETNs



Three Ways Exchange-Traded Products Create Commodity Exposure

Structure	What you actually own/expose yourself to	Example exposure
Physical-backed	Fund/trust holds the commodity	Gold
Futures-based	Fund continually owns and rolls futures	Oil, gas, agriculture
ETN	Bank debt whose payoff references a commodity/index	Commodity indexes

BEFORE BUYING, ASK: WHAT DOES THE PRODUCT ACTUALLY HOLD?

Physical → Futures → Swaps → Commodity stocks → Index?

Commodity Producer Stocks: Indirect Commodity Exposure



HOW COMMODITY PRICES REACH EQUITY RETURNS



PRODUCER EQUITY RETURN DEPENDS ON

PRIMARY DRIVERS

- + Commodity price
- + Production volume

COST & CAPITAL DRAGS

- Operating costs
- Capital spending
- Debt / financing costs

ADJUSTMENTS

- ± Hedging
- ± Company-specific / political risk

Producer equity return depends on more than the commodity price alone.

Forwards & Swaps:

Customized OTC Commodity Hedges



HOW THE INSTRUMENTS DIFFER

	Futures	Forward	Swap
Trading venue	Exchange	OTC	OTC / sometimes cleared
Terms	Standardized	Customized	Customized
Typical settlement	Cash or physical	Often physical	Usually financial
Maturity	Exchange months	Any agreed date	Multiple periods
Counterparty	Clearinghouse	Specific counterparty	Specific counterparty/dealer
Main use	Hedge / speculate	Lock physical transaction price	Hedge recurring price exposure

FORWARD

Agree today on the price and terms of a future physical transaction.

SWAP

Exchange a floating commodity price for a fixed price over one or more periods.

Commodity-Specific Risks: Basis, Curves & Spreads



A COMMODITY POSITION HAS MORE THAN ONE PRICE RISK

01	FLAT PRICE RISK	Will the commodity itself rise or fall?
02	BASIS RISK	Will my physical price move differently from my hedge?
03	CURVE RISK	Will different contract months move differently?
04	SPREAD RISK	Will location, quality, or calendar differentials change?
05	ROLL RISK	What happens when I replace an expiring futures contract?

TERM STRUCTURE EXAMPLE

Today		Oct		Nov		Dec		Jan
\$80	→	\$79	→	\$77	→	\$76	→	\$75
Spot		Futures		Futures		Futures		Futures

One Commodity Thesis, Five Ways to Trade It

SAME THESIS → DIFFERENT TRADES

Instrument	Example trade	What you are really betting on
Futures	Long WTI future	Oil price rises
Call option	Buy \$80 WTI call	Oil rises enough, soon enough
Commodity ETP	Buy oil futures-based fund	Oil futures + roll structure
Producer stock	Buy oil producer	Oil rises + company converts it into profit
Forward / swap	Fix future oil price	Commercial price exposure

One Commodity Thesis, Five Ways to Trade It

WHAT EXACTLY AM I TRYING TO EXPRESS OR HEDGE?

01 Do I have a directional commodity view? **YES**

02 Do I want direct or indirect exposure?
DIRECT PRICE → Futures / Options / Commodity ETP
COMPANY → Producer stock
COMMERCIAL HEDGE → Futures / Forward / Swap

THEN NARROW FURTHER

03 Do I want linear exposure? **YES** → Futures **NO, I WANT DEFINED DOWNSIDE** → Options

04 Do I want simple retail access?
YES → Commodity ETP
NO / COMFORTABLE WITH DERIVATIVES → Futures or options

05 Do I have a specific physical exposure that does not match an exchange contract?
YES → Forward / Swap

Questions